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Java SE 27 & JDK 27

Class DeflaterOutputStream

java.lang.Object
  java.io.OutputStream
    java.io.FilterOutputStream
      java.util.zip.DeflaterOutputStream

All Implemented Interfaces:

`Closeable`, `Flushable`, `AutoCloseable`

Direct Known Subclasses:

`GZIPOutputStream`, `ZipOutputStream`

```
public class DeflaterOutputStream
extends FilterOutputStream
```

This class implements an output stream filter for compressing data in the "deflate" compression format. It is also used as the basis for other types of compression filters, such as GZIPOutputStream.

Unless otherwise noted, passing a `null` argument to a constructor or method in this class will cause a `NullPointerException` to be thrown.

Compressor Usage

A `DeflaterOutputStream` created without specifying a compressor will create a compressor at construction time, and close the compressor when the output stream is closed.

If a compressor is specified when creating a `DeflaterOutputStream`, it is the responsibility of the caller to close the compressor after closing the output stream.

API Note:

The `close()` method should be called to release resources used by this stream, either directly, or with the `try`-with-resources statement.

Since:

1.1

See Also:

`Deflater`

Field Summary

Fields

Modifier and Type	Field	Description

protected byte[]	**buf**	Output buffer for writing compressed data.
protected Deflater	**def**	Compressor for this stream.

Fields declared in class FilterOutputStream

out

Constructor Summary

Constructors

Constructor	Description
DeflaterOutputStream (OutputStream out)	Creates a new output stream and compressor with the default compression level and a default buffer size.
DeflaterOutputStream (OutputStream out, boolean syncFlush)	Creates a new output stream and compressor with the default compression level, a default buffer size and the specified flush mode.
DeflaterOutputStream (OutputStream out, Deflater def)	Creates a new output stream with the specified compressor and a default buffer size.
DeflaterOutputStream (OutputStream out, Deflater def, boolean syncFlush)	Creates a new output stream with the specified compressor, flush mode and a default buffer size.
DeflaterOutputStream (OutputStream out, Deflater def, int size)	Creates a new output stream with the specified compressor and buffer size.
DeflaterOutputStream (OutputStream out, Deflater def, int size, boolean syncFlush)	Creates a new output stream with the specified compressor, buffer size and flush mode.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
void	close()	Writes remaining compressed data to the output stream and closes the underlying stream.
protected void	deflate()	Writes next block of compressed data to the output stream.
void	finish()	Finishes writing compressed data to the output stream without closing the underlying stream.
void	flush()	Flushes the compressed output stream.
void	write(byte[] b, int off, int len)	Writes an array of bytes to the compressed output stream.
void	write(int b)	Writes a byte to the compressed output stream.

Methods declared in class FilterOutputStream

write

Methods declared in class OutputStream

nullOutputStream

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Details

def

```
protected Deflater def
```

Compressor for this stream.

buf

```
protected byte[] buf
```

Output buffer for writing compressed data.

Constructor Details

DeflaterOutputStream

```
public DeflaterOutputStream(OutputStream out,
                            Deflater def,
                            int size,
                            boolean syncFlush)
```

Creates a new output stream with the specified compressor, buffer size and flush mode.

Closing this output stream will not close the given compressor.

Parameters:

`out` - the output stream

`def` - the compressor ("deflater")

`size` - the output buffer size

`syncFlush` - if `true` the `flush()` method of this instance flushes the compressor with flush mode `Deflater.SYNC_FLUSH` before flushing the output stream, otherwise only flushes the output stream

Throws:

`IllegalArgumentException` - if `size <= 0`

Since:

1.7

DeflaterOutputStream

```
public DeflaterOutputStream(OutputStream out,
                            Deflater def,
                            int size)
```

Creates a new output stream with the specified compressor and buffer size.

The new output stream instance is created as if by invoking the 4-argument constructor `DeflaterOutputStream(out, def, size, false)`.

Closing this output stream will not close the given compressor.

Parameters:

`out` - the output stream

`def` - the compressor ("deflater")

`size` - the output buffer size

Throws:

`IllegalArgumentException` - if `size <= 0`

DeflaterOutputStream

```
public DeflaterOutputStream(OutputStream out,
                            Deflater def,
                            boolean syncFlush)
```

Creates a new output stream with the specified compressor, flush mode and a default buffer size.

Closing this output stream will not close the given compressor.

Parameters:

`out` - the output stream

`def` - the compressor ("deflater")

`syncFlush` - if `true` the `flush()` method of this instance flushes the compressor with flush mode `Deflater.SYNC_FLUSH` before flushing the output stream, otherwise only flushes the output stream

Since:

1.7

DeflaterOutputStream

```
public DeflaterOutputStream(OutputStream out,
                            Deflater def)
```

Creates a new output stream with the specified compressor and a default buffer size.

The new output stream instance is created as if by invoking the 3-argument constructor `DeflaterOutputStream(out, def, false)`.

Closing this output stream will not close the given compressor.

Parameters:

`out` - the output stream

`def` - the compressor ("deflater")

DeflaterOutputStream

```
public DeflaterOutputStream(OutputStream out,
                            boolean syncFlush)
```

Creates a new output stream and compressor with the default compression level, a default buffer size and the specified flush mode.

The compressor will be closed when this output stream is closed.

Parameters:

`out` - the output stream

`syncFlush` - if `true` the `flush()` method of this instance flushes the compressor with flush mode `Deflater.SYNC_FLUSH` before flushing the output stream, otherwise only flushes the output stream

Since:

1.7

DeflaterOutputStream

```
public DeflaterOutputStream(OutputStream out)
```

Creates a new output stream and compressor with the default compression level and a default buffer size.

The new output stream instance is created as if by invoking the 2-argument constructor `DeflaterOutputStream(out, false)`.

The compressor will be closed when this output stream is closed.

Parameters:

`out` - the output stream

Method Details

write

```
public void write(int b)
           throws IOException
```

Writes a byte to the compressed output stream. This method will block until the byte can be written.

Overrides:

`write` in class `FilterOutputStream`

Parameters:

`b` - the byte to be written

Throws:

`IOException` - if an I/O error has occurred

write

```
public void write(byte[] b,
                  int off,
                  int len)
           throws IOException
```

Writes an array of bytes to the compressed output stream. This method will block until all the bytes are written.

Overrides:

`write` in class `FilterOutputStream`

Parameters:

`b` - the data to be written

`off` - the start offset of the data

`len` - the length of the data

Throws:

`IOException` - if an I/O error has occurred

See Also:

`FilterOutputStream.write(int)`

finish

```
public void finish()
            throws IOException
```

Finishes writing compressed data to the output stream without closing the underlying stream. Use this method when applying multiple filters in succession to the same output stream.

Throws:

`IOException` - if an I/O error has occurred

close

```
public void close()
           throws IOException
```

Writes remaining compressed data to the output stream and closes the underlying stream.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Overrides:

`close` in class `FilterOutputStream`

Throws:

`IOException` - if an I/O error has occurred

See Also:

`FilterOutputStream.flush()`, `FilterOutputStream.out`

deflate

```
protected void deflate()
               throws IOException
```

Writes next block of compressed data to the output stream.

Throws:

`IOException` - if an I/O error has occurred

flush

```
public void flush()
           throws IOException
```

Flushes the compressed output stream. If `syncFlush` is `true` when this compressed output stream is constructed, this method first flushes the underlying `compressor` with the flush mode `Deflater.SYNC_FLUSH` to force all pending data to be flushed out to the output stream and then flushes the

output stream. Otherwise this method only flushes the output stream without flushing the `compressor`.

Specified by:

`flush` in interface `Flushable`

Overrides:

`flush` in class `FilterOutputStream`

Throws:

`IOException` - if an I/O error has occurred

Since:

1.7

See Also:

`FilterOutputStream.out`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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